UTS Sistem Basis Data

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6.10 Disini kita akan membuat table Hotel dengan integrity enhancement features

```
uts_basis_data=# create extension if not exists btree_gist;
CREATE EXTENSION
```

```
uts_basis_data=# create table hotel (
uts_basis_data(# hotelNo varchar(15) primary key,
uts_basis_data(# hotelName varchar(40) not null,
uts_basis_data(# city varchar(40) not null
uts_basis_data(# );
```

6.11 Membuat table Room, Booking, Guest dengan catatan

(a) type must be one of Single, Double, or Family.
(b) price must be between £10 and £100.
(c) roomNo must be between 1 and 100.
(d) dateFrom and dateTo must be greater than today's date.

(e) The same room cannot be double-booked.
(f) The same guest cannot have overlapping bookings.

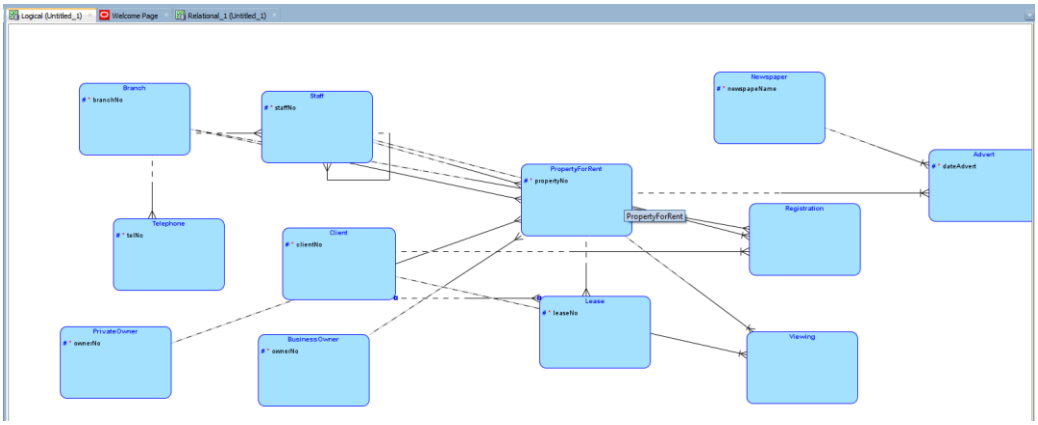
```
uts_basis_data=# create table room (
uts_basis_data(# roomNo int not null,
uts_basis_data(# hotelNo varchar(15) not null,
uts_basis_data(# type varchar(15) not null,
uts_basis_data(# price decimal(5,2) not null,
uts_basis_data(# primary key (roomNo, hotelNo),
uts_basis_data(# foreign key (hotelNo) references hotel(hotelNo) on delete cascade,
uts_basis_data(# constraint chk_room_type check (type in ('Single', 'Double', 'Family')),
uts_basis_data(# constraint chk_room_price check (price between 10 and 100),
uts_basis_data(# constraint chk_room_no check (roomNo between 1 and 100)
uts_basis_data(# );
CREATE TABLE
```

```
uts_basis_data=# create table guest (
uts_basis_data(# guestNo varchar(15) primary key,
uts_basis_data(# guestName varchar(40) not null,
uts_basis_data(# guestAddress varchar(110) not null
uts_basis_data(# );
CREATE TABLE
```

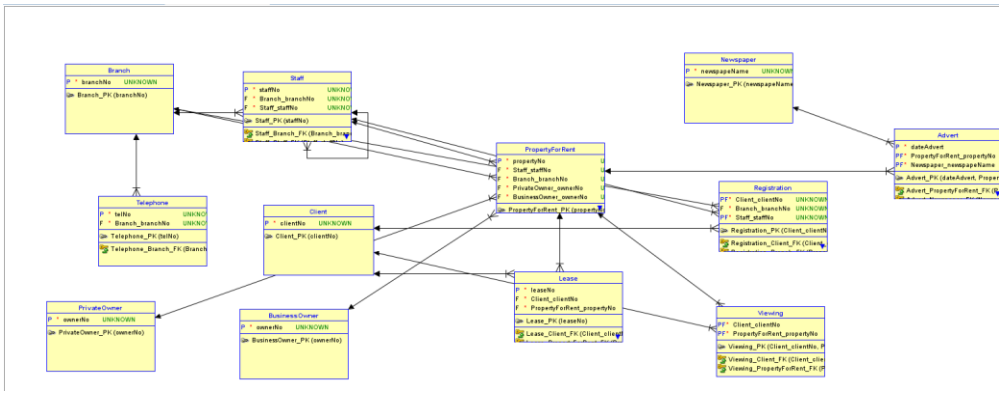
```
uts_basis_data=#  create table booking (
uts_basis_data(# hotelNo varchar(15) not null,
uts_basis_data(# guestNo varchar(15) not null,
uts_basis_data(# dateFrom date not null,
uts_basis_data(# dateTo date not null,
uts_basis_data(# roomNo int not null,
uts_basis_data(# primary key (hotelNo, guestNo, dateFrom),
uts_basis_data(# foreign key (hotelNo) references hotel(hotelNo) on delete cascade,
uts_basis_data(# foreign key (guestNo) references guest(guestNo) on delete cascade,
uts_basis_data(# foreign key (roomNo, hotelNo) references room(roomNo, hotelNo) on delete cascade,
uts_basis_data(# constraint chk_dates_future check (dateFrom > current_date and dateTo > current_date),
uts_basis_data(# constraint chk_date_logic check (dateTo > dateFrom),
uts_basis_data(# constraint exclude_room_overlap exclude using gist (
uts_basis_data(# roomNo with =,
uts_basis_data(# hotelNo with =,
uts_basis_data(# daterange(dateFrom, dateTo) with &&
uts_basis_data(# ),
uts_basis_data(# constraint exclude_guest_overlap exclude using gist (
uts_basis_data(# guestNo with =,
uts_basis_data(# daterange(dateFrom, dateTo) with &&
uts_basis_data(# )
uts_basis_data(# );
```

16.1 Membuat Logical Model kemudian diubah menjadi physical model dan generate DDL kemudian DUMP di postgres

Logical Model



Physical Model



DDL dalam query Oracle 11g belum diubah ke query Postgres

```
CREATE TABLE Advert
    (
     dateAdvert                 UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     PropertyForRent_propertyNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     Newspaper_newspapeName     UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE Advert
    ADD CONSTRAINT Advert_PK PRIMARY KEY ( dateAdvert, PropertyForRent_propertyNo, Newspaper_newspapeName ) ;

CREATE TABLE Branch
    (
     branchNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE Branch
    ADD CONSTRAINT Branch_PK PRIMARY KEY ( branchNo ) ;

CREATE TABLE BusinessOwner
    (
     ownerNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE BusinessOwner
    ADD CONSTRAINT BusinessOwner_PK PRIMARY KEY ( ownerNo ) ;

CREATE TABLE Client
```

```
CREATE TABLE Client
    (
     clientNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE Client
    ADD CONSTRAINT Client_PK PRIMARY KEY ( clientNo ) ;

CREATE TABLE Lease
    (
     leaseNo                    UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     Client_clientNo            UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     PropertyForRent_propertyNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE Lease
    ADD CONSTRAINT Lease_PK PRIMARY KEY ( leaseNo ) ;

CREATE TABLE Newspaper
    (
     newspapeName UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE Newspaper
    ADD CONSTRAINT Newspaper_PK PRIMARY KEY ( newspapeName ) ;

CREATE TABLE PrivateOwner
    (
     ownerNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE PrivateOwner
    ADD CONSTRAINT PrivateOwner_PK PRIMARY KEY ( ownerNo ) ;

CREATE TABLE PropertyForRent
```

```
CREATE TABLE PropertyForRent
    (
     propertyNo             UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     Staff_staffNo          UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     Branch_branchNo        UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     PrivateOwner_ownerNo   UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     BusinessOwner_ownerNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE PropertyForRent
    ADD CONSTRAINT PropertyForRent_PK PRIMARY KEY ( propertyNo ) ;

CREATE TABLE Registration
    (
     Client_clientNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     Branch_branchNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     Staff_staffNo   UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE Registration
    ADD CONSTRAINT Registration_PK PRIMARY KEY ( Client_clientNo, Staff_staffNo ) ;

CREATE TABLE Staff
    (
     staffNo        UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     Branch_branchNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     Staff_staffNo   UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
```

```
ALTER TABLE Staff
    ADD CONSTRAINT Staff_PK PRIMARY KEY ( staffNo ) ;

CREATE TABLE Telephone
    (
     telNo          UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     Branch_branchNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE Telephone
    ADD CONSTRAINT Telephone_PK PRIMARY KEY ( telNo ) ;

CREATE TABLE Viewing
    (
     Client_clientNo            UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL ,
     PropertyForRent_propertyNo UNKNOWN
--  ERROR: Datatype UNKNOWN is not allowed
                    NOT NULL
    )
;

ALTER TABLE Viewing
    ADD CONSTRAINT Viewing_PK PRIMARY KEY ( Client_clientNo, PropertyForRent_propertyNo ) ;

ALTER TABLE Advert
    ADD CONSTRAINT Advert_Newspaper_FK FOREIGN KEY
    (
     Newspaper_newspapeName
    )
    REFERENCES Newspaper
    (
     newspapeName
    )
;

ALTER TABLE Advert
    ADD CONSTRAINT Advert_PropertyForRent_FK FOREIGN KEY
    (
     PropertyForRent_propertyNo
    )
    REFERENCES PropertyForRent
    (
     propertyNo
```

```
ALTER TABLE Lease
    ADD CONSTRAINT Lease_Client_FK FOREIGN KEY
    (
     Client_clientNo
    )
    REFERENCES Client
    (
     clientNo
    )
;

ALTER TABLE Lease
    ADD CONSTRAINT Lease_PropertyForRent_FK FOREIGN KEY
    (
     PropertyForRent_propertyNo
    )
    REFERENCES PropertyForRent
    (
     propertyNo
    )
;

ALTER TABLE PropertyForRent
    ADD CONSTRAINT PropertyForRent_Branch_FK FOREIGN KEY
    (
     Branch_branchNo
    )
    REFERENCES Branch
    (
     branchNo
    )
;

--  ERROR: FK name length exceeds maximum allowed length(30)
ALTER TABLE PropertyForRent
    ADD CONSTRAINT PropertyForRent_BusinessOwner_FK FOREIGN KEY
    (
     BusinessOwner_ownerNo
    )
    REFERENCES BusinessOwner
    (
     ownerNo
    )
;

--  ERROR: FK name length exceeds maximum allowed length(30)
ALTER TABLE PropertyForRent
```

```
--  ERROR: FK name length exceeds maximum allowed length(30)
ALTER TABLE PropertyForRent
    ADD CONSTRAINT PropertyForRent_PrivateOwner_FK FOREIGN KEY
    (
     PrivateOwner_ownerNo
    )
    REFERENCES PrivateOwner
    (
     ownerNo
    )
;

ALTER TABLE PropertyForRent
    ADD CONSTRAINT PropertyForRent_Staff_FK FOREIGN KEY
    (
     Staff_staffNo
    )
    REFERENCES Staff
    (
     staffNo
    )
;

ALTER TABLE Registration
    ADD CONSTRAINT Registration_Branch_FK FOREIGN KEY
    (
     Branch_branchNo
    )
    REFERENCES Branch
    (
     branchNo
    )
;

ALTER TABLE Registration
    ADD CONSTRAINT Registration_Client_FK FOREIGN KEY
    (
     Client_clientNo
    )
    REFERENCES Client
    (
     clientNo
    )
;

ALTER TABLE Registration
    ADD CONSTRAINT Registration_Staff_FK FOREIGN KEY
    (
     Staff_staffNo
    )
    REFERENCES Staff
    (
```

```
ALTER TABLE Staff
    ADD CONSTRAINT Staff_Branch_FK FOREIGN KEY
    (
     Branch_branchNo
    )
    REFERENCES Branch
    (
     branchNo
    )
;

ALTER TABLE Staff
    ADD CONSTRAINT Staff_Staff_FK FOREIGN KEY
    (
     Staff_staffNo
    )
    REFERENCES Staff
    (
     staffNo
    )
;

ALTER TABLE Telephone
    ADD CONSTRAINT Telephone_Branch_FK FOREIGN KEY
    (
     Branch_branchNo
    )
    REFERENCES Branch
    (
     branchNo
    )
;

ALTER TABLE Viewing
    ADD CONSTRAINT Viewing_Client_FK FOREIGN KEY
    (
     Client_clientNo
    )
    REFERENCES Client
    (
     clientNo
    )
;

ALTER TABLE Viewing
    ADD CONSTRAINT Viewing_PropertyForRent_FK FOREIGN KEY
    (
     PropertyForRent_propertyNo
    )
    REFERENCES PropertyForRent
```

```
ALTER TABLE Viewing
    ADD CONSTRAINT Viewing_PropertyForRent_FK FOREIGN KEY
    (
     PropertyForRent_propertyNo
    )
    REFERENCES PropertyForRent
    (
     propertyNo
    )
;
```